



STEVENS
INSTITUTE *of* TECHNOLOGY
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Data Warehousing and Business Intelligence in Retail Banking

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Overview

This case study will focus on an unnamed regional chain of retail banks, referred to as Alpha Bank, operating primarily in the Northeast Region and Mid-Atlantic states. This mid-sized financial services corporation boasts around 120 branches across six states and an annual revenue of approximately \$800 million. The executive team's current objective for the next few years is to expand operations into the Southeast Region and major metropolitan areas, such as Miami and Atlanta, while also launching innovative digital banking solutions tailored to younger demographics, particularly Gen-Z and young professionals. However, Alpha Bank still grapples with an antiquated legacy BI system, utilizing Oracle as the backend database. This system suffers from disparate reporting structures and a lack of data integration and understanding across different departments of the organization.

Opportunity Matrix

Business Process / Event	Customer Service	Human Resources	Organization/Workgroup Data Science/IT	Risk Management	Marketing
Credit analysis				X	X
High-value customer identification	X				X
Loan default prediction			X	X	
Employee support/training		X	X		
Smart offer development				X	X

High-level Bus Matrix

Business Process / Event	Date	Customer	Loan	Bank Account	Credit Card _{Common Dimensions}	Branch	Employee	Offer Type
Credit analysis	X	X	X	X	X			X

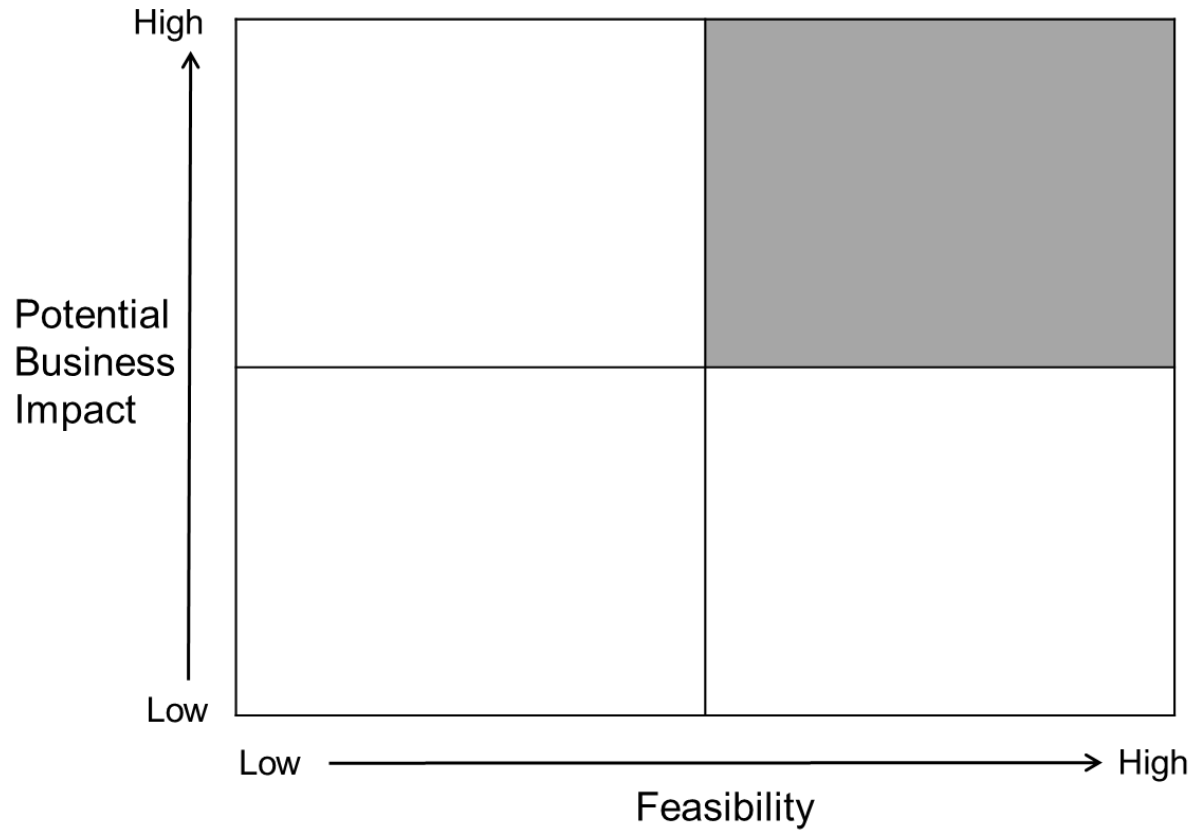
High-value customer identification	X					X		
Loan default prediction	X							
Employee support/training	X					X	X	
Smart offer development	X	X	X		X			X

Prioritization Grid

Loan
default
prediction

Credit
Analysis

Employee
support/tra
ining



developme nt	Smart offer
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High-value customer

identification

Detailed Bus Matrix

Business Process / Event	Fact Tables	Granularity	Facts	Date	Customer	Credit Card	Offer Type	Bank Account
Loan default prediction	Loan payment transactions	1 row for every loan payment transaction	Loan payment amount	X	X			X
	Checking account transaction	1 row for every checking account transaction	Checking account transaction amount	X	X			X
Smart offer development	Credit card applications	1 row for every credit card application	Applicant chosen terms Applicant salary	X	X	X	X	X
	Loan applications	1 row for every loan application	Applicant chosen terms Applicant salary	X	X		X	X
	Offer types	1 row per offer APR and terms previously offered by the bank	Offer APR Offer term Offer premium Offer initial payment	X		X	X	

Conformed Dimension

1. Standardization:

- The initial step involves standardizing the data coding methods and formats across different data sources from various branches. This ensures uniformity and consistency in the data, providing a solid groundwork for subsequent data matching processes.

2. Matching and Deduplication:

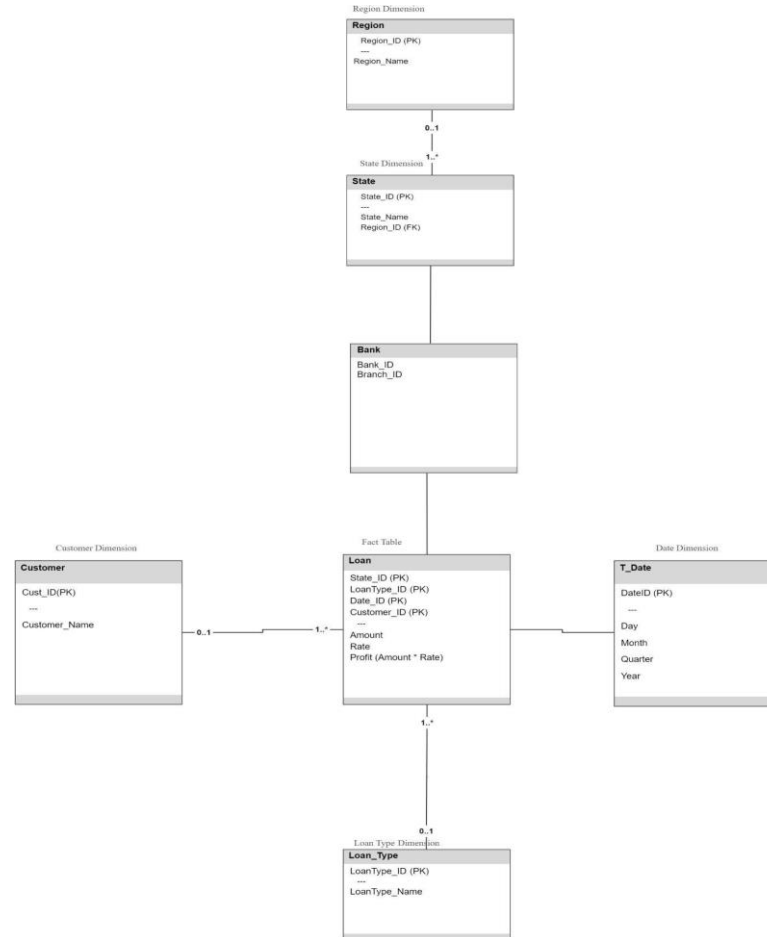
- This phase encompasses two key aspects of data matching. Firstly, it involves matching the identifiers and attributes of the same entities from different data sources to enhance data accuracy and completeness. Secondly, it focuses on identifying duplicate records within the datasets, which serves as a crucial foundation for the subsequent filtering step.

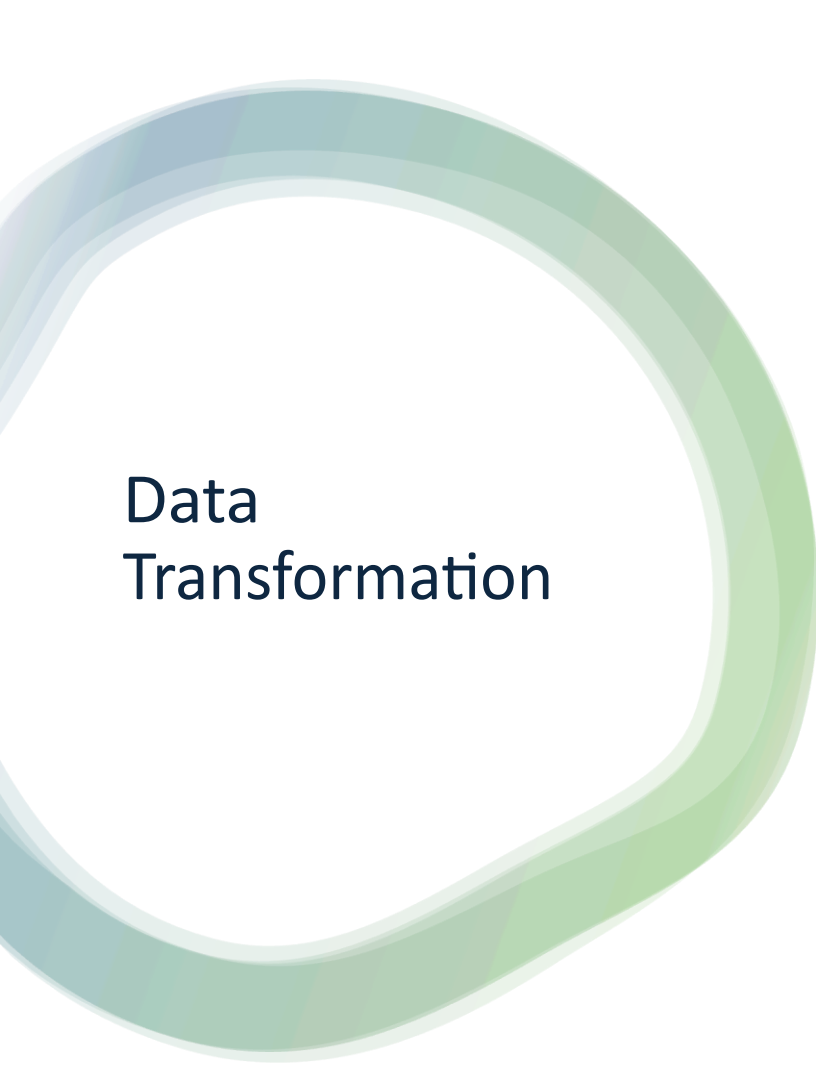
3. Filtering:

- The primary objective of data filtering is to identify and select the conformed dimensions as the master data. This involves evaluating the matched data to determine the most accurate and reliable dimension data for further processing.
- Possible Conformed Dimensions:
 - - Date Dimension
 - - Customer Dimension
 - - Loan Dimension
 - - Credit Card Dimension
 - - Branch Dimension

- These steps ensure that the conformed dimensions are effectively identified, standardized, matched, and filtered to facilitate accurate and efficient data integration and analysis across the organization's various branches and data sources.

Dimensional Model





Data Transformation

Post-extraction, data undergoes transformation, including standardizing types, correcting errors, and formatting. This step, the most complex and time-consuming in ETL, ranges from basic conversions to advanced cleansing.

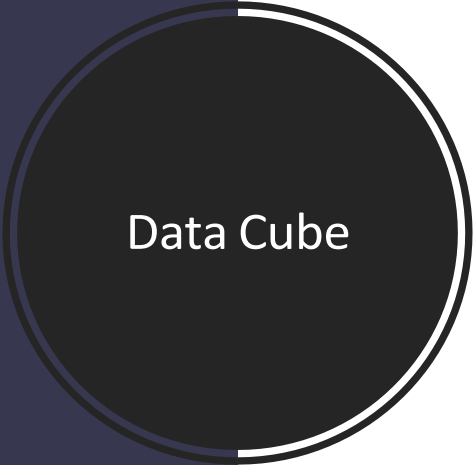
Transformation Rules:

1. Selective Loading: Only relevant columns are loaded for efficiency.
2. Standardized Coding: Ensures uniformity and compatibility.
3. Pre-calculation: Compute values to streamline processing.
4. Reordering: Optimize queries by rearranging data.
5. Integration and Deduplication: Merge sources and remove duplicates.
6. Row/Column Transformation: Modify data structures for analysis.
7. Unpivot/Pivot: Convert data formats as needed.
8. Pre-Connection: Prepare data for downstream systems.
9. Data Validation: Verify data meets standards for quality and reliability.

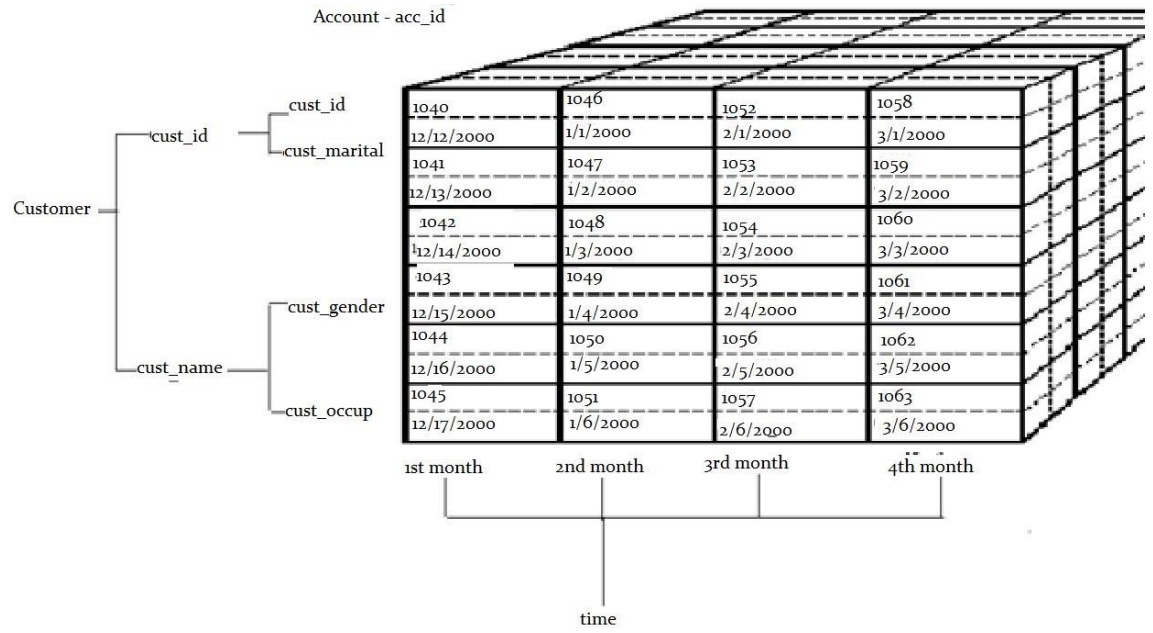
These rules ensure efficient processing, quality data, and readiness for analysis.

Sample Two-Way Aggregate Table

time_key		cust_occupation_key	count	total_revenue
546512		1006	6,363	\$9,544,936.25
546512		1009	1,758	\$11,504,352.78
546512		1011	353	\$4,577,192.15
546118		1004	9,098	\$10,924,696.40
546118		1003	23,544	\$13,843,872.34
546087		1010	5,409	\$7,677,899.22



Data Cube



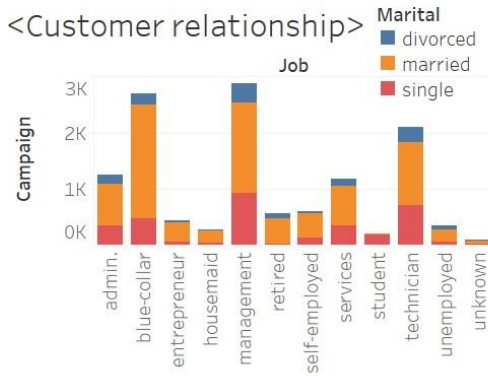
User/Task
Analysis

User Role	Task	Delivery
Executives/Upper Management	Access high-level summaries to inform business strategy	Portal, reports
Knowledge workers: risk management/compliance staff, Data Science/quants	Ad-hoc queries for risk analysis, predictive modeling, revenue forecasting, etc	Portal, OLAP, MOLAP
Operational Workers: personal bankers, HR	Access to customer/employee data for everyday client-facing or administrative purposes	Portal, operational reports

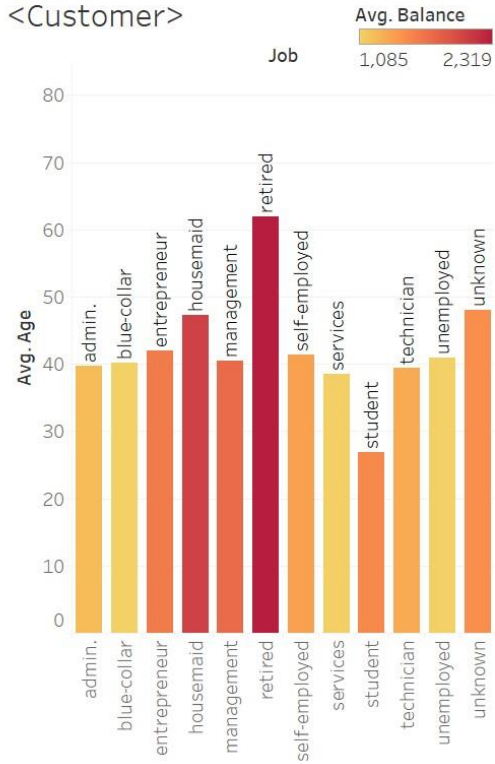


Example
visualization:
Customer
demographic
dashboard in use by
the marketing team

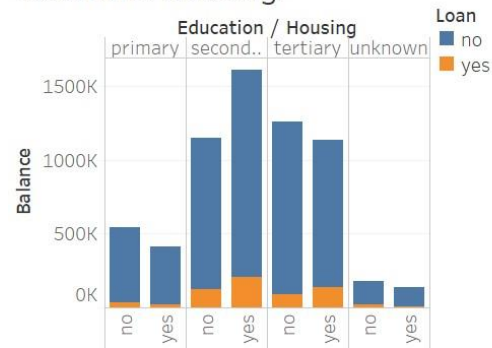
<Customer relationship>



<Customer>



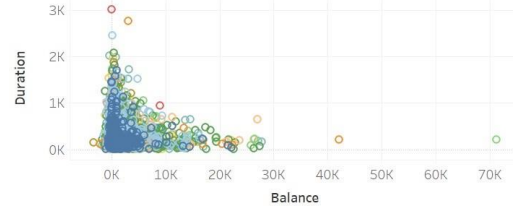
<Loan and Housing>



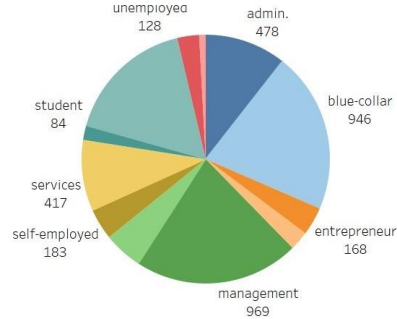


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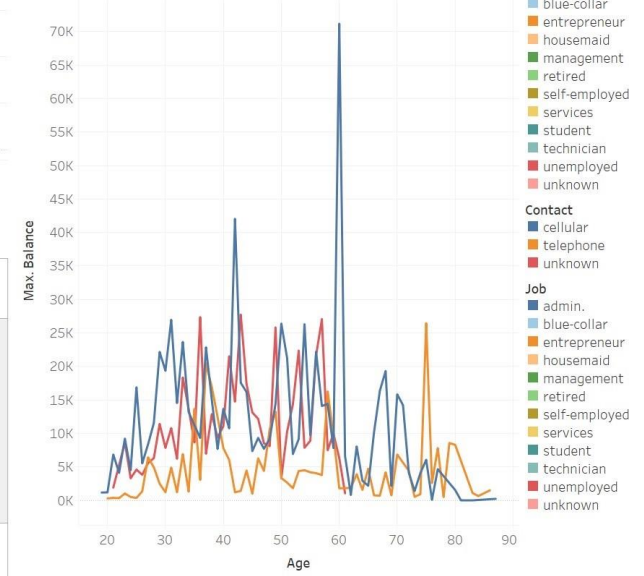
<Customer target>



<Customer Overview>

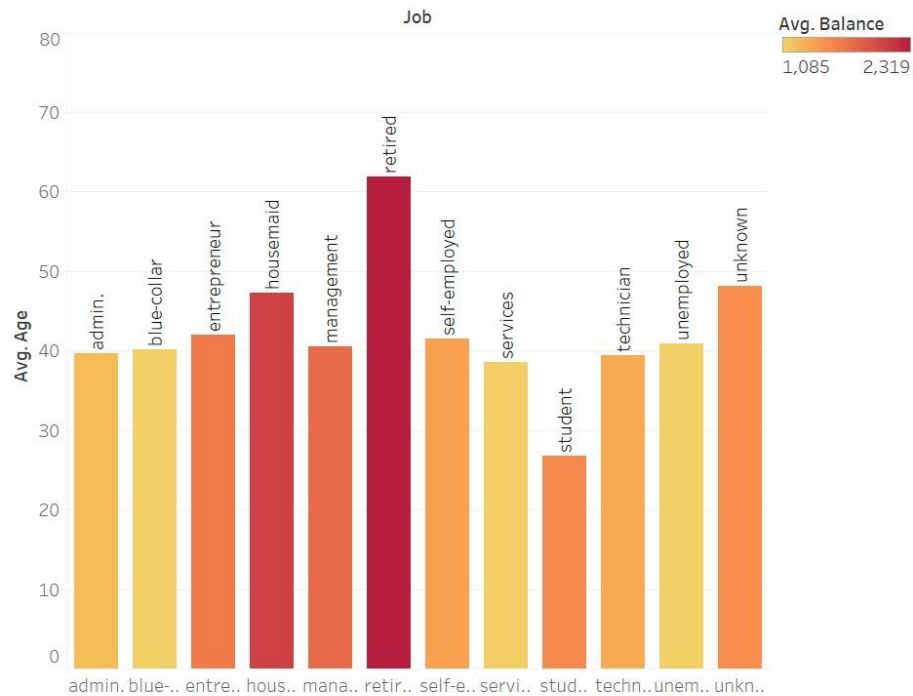


<Contact type>



Customer
Insights

<Customer>

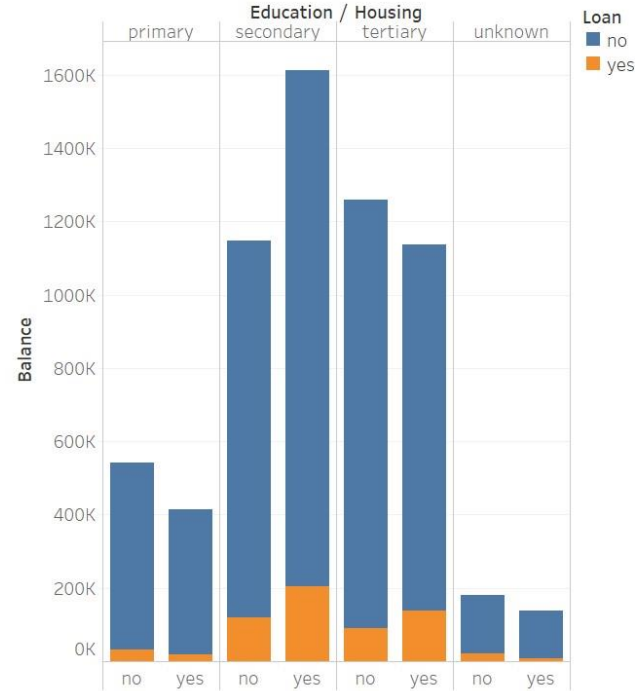


Average of Age for each Job. Color shows average of Balance. The marks are labeled by Job.



Education
and Financial
status

<Loan and Housing>

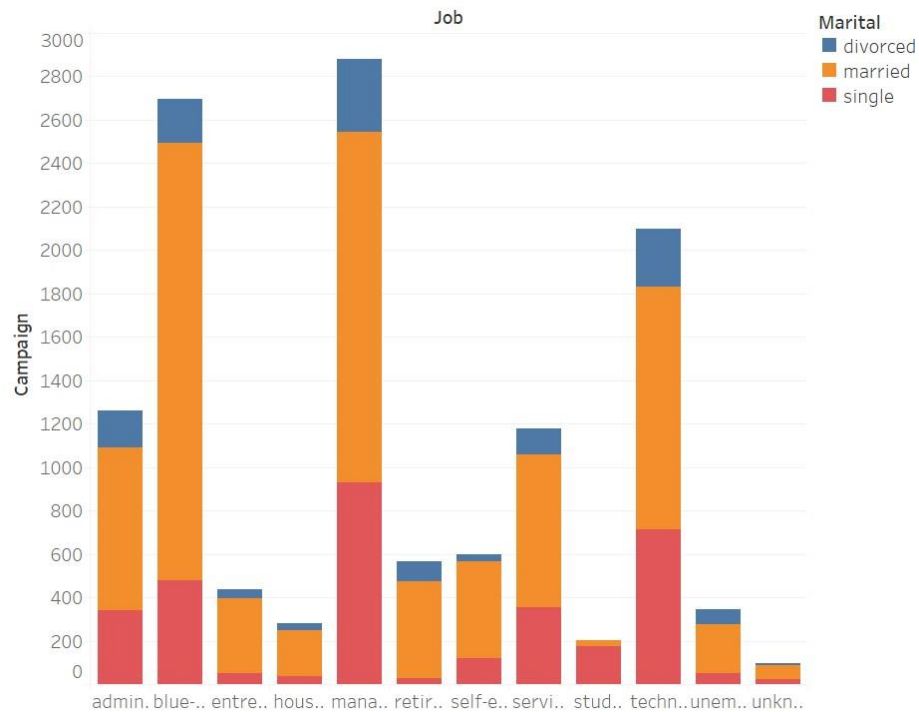


Sum of Balance for each Housing broken down by Education. Color shows details about Loan.



Campaign
with job and
marital status

<Customer relationship>

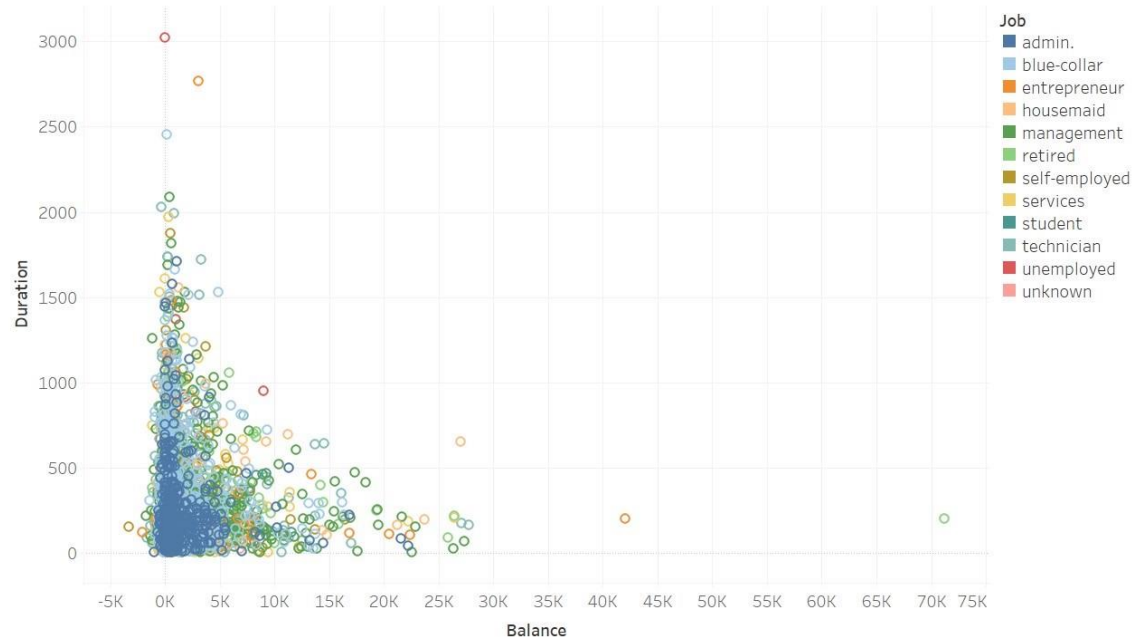


Sum of Campaign for each Job. Color shows details about Marital.

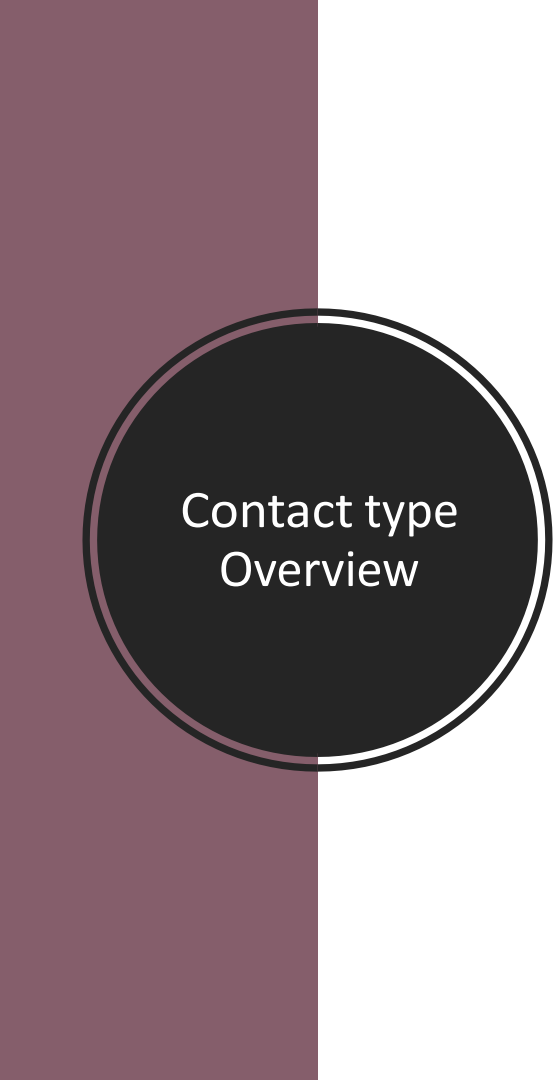


Target Insights

<Customer target>

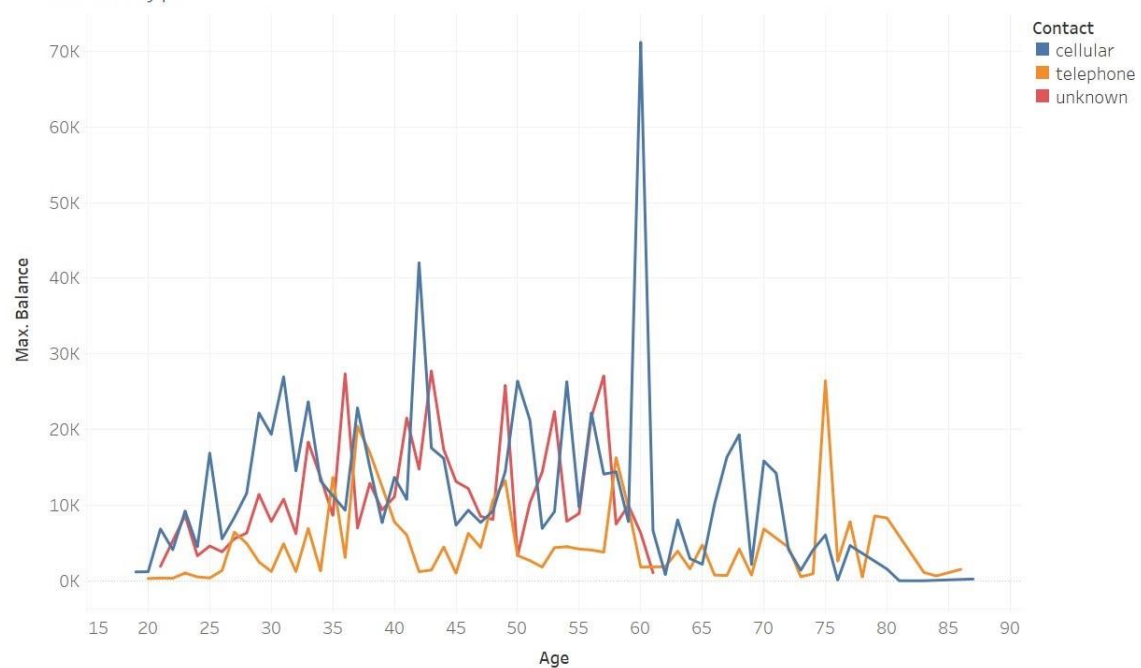


Balance vs. Duration. Color shows details about Job.



Contact type Overview

<Contact type>

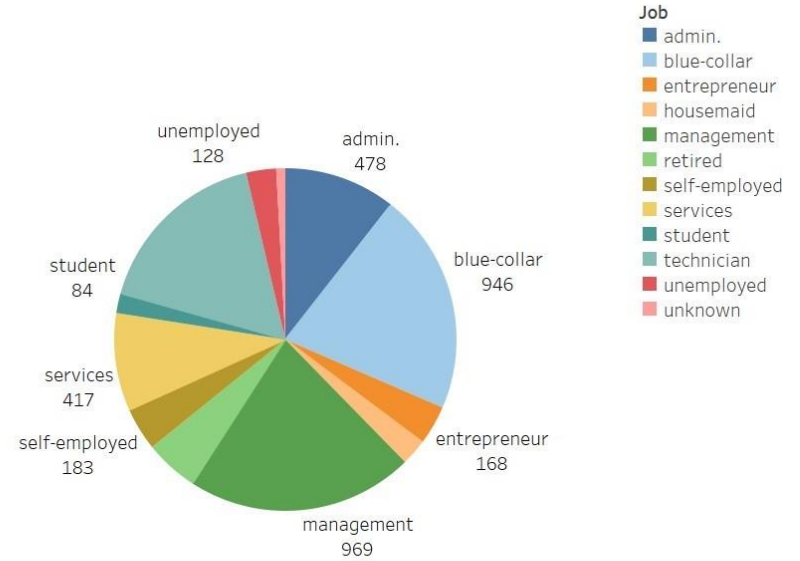


The trend of maximum of Balance for Age. Color shows details about Contact.



Customer Overview

<Customer Overview>



Job and count of Job. Color shows details about Job. The marks are labeled by Job and count of Job.

Thank you